

An algebraic version of Herbrand's theorem

You can find the recording of the talk at <https://youtu.be/H70XZcU1kkQ>

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Based on a joint work with Francesca Guffanti:



Freely adding one layer of quantifiers to a Boolean doctrine. *ArXiv link*.

Universal theory: universal closures of quantifier-free formulas.

E.g.: theory of partial orders.

1. (Reflexivity) $\forall x (x \leq x)$,
2. (Transitivity) $\forall x \forall y \forall z ((x \leq y \wedge y \leq z) \rightarrow x \leq z)$
3. (Anti-symmetry) $\forall x \forall y ((x \leq y) \wedge (y \leq x) \rightarrow x = y)$

Also, the theory of partial orders with $\min$ and $\max$ (as constants).

4. $\forall x (\min \leq x)$,
5. $\forall x (x \leq \max)$.

Herbrand's theorem for $\exists$ -statements, 1930

Let $\mathcal{T}$ be a universal theory in a language with at least a constant symbol. For $x \in \text{Var}$ and $\alpha(x)$ quantifier-free, $\mathcal{T}$ proves $\exists x \alpha(x)$ if and only if there are term-definable constants (i.e., ground terms) $c_1, \dots, c_n$ such that $\mathcal{T}$ proves $\alpha[c_1/x] \vee \dots \vee \alpha[c_n/x]$.

Examples:

- ($\Leftarrow$) The theory $\mathcal{T}$ of partial orders with min and max proves

$$\exists x (\max \leq x)$$

because $\mathcal{T}$ proves

$$\max \leq \max.$$

- ($\Rightarrow$) The theory $\mathcal{T}$ of partial orders with min and max does not prove

$$\exists x (\min \not\leq x \not\leq \max)$$

because $\mathcal{T}$ does not prove any of the following two:

$$\min \not\leq \min \not\leq \max \qquad \min \not\leq \max \not\leq \max,$$

as well as any disjunction made up of them.

Herbrand's theorem

Let $\mathcal{T}$ be a universal theory *in a language with at least a constant symbol*. For $x \in \text{Var}$ and $\alpha(x)$ quantifier-free, $\mathcal{T}$ proves $\exists x \alpha(x)$ if and only if there are term-definable constants $c_1, \dots, c_n$ such that $\mathcal{T}$ proves $\alpha[c_1/x] \vee \dots \vee \alpha[c_n/x]$.

The hypothesis that there is at least a constant symbol cannot be removed: $\vdash \exists x (x = x)$, but no witnessing constants.

However, the hypothesis can be removed if we either

1. replace “term-definable constants” by “terms” (so that also variables can be used),
2. use a version of classical first-order logic whose semantics admits empty structures.

Boolean algebras : Classical propositional logic

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First-order Boolean doctrines [Lawvere, '60s] : Classical first-order logic.

Fix a functional language $\mathcal{F}$.

Roughly speaking: first-order Boolean doctrines over $\mathcal{F}$ are the algebras of the form

$$(\mathcal{F}, \mathcal{R})\text{-Formulas}/\equiv_{\mathcal{T}}$$

for $\mathcal{R}$ ranging among all sets of relation symbols, and $\mathcal{T}$ among all theories in the first-order language $(\mathcal{F}, \mathcal{R})$.

Usually, first-order Boolean doctrines are defined as certain functors $\mathbf{C}^{\text{op}} \rightarrow \mathbf{BoolAlg}$, with $\mathbf{C}$ taking the place of $\mathcal{F}$.

We avoid this categorical phrasing in this talk.

A *first-order Boolean doctrine* over a functional language $\mathcal{F}$ consists of

1. **a family of Boolean algebras** $(\mathbf{P}(X))_{X \subseteq_{\text{fin}} \text{Var}}$ (here, $\mathbf{P}(X)$ models the set of equivalence classes of formulas with free variables in X),
2. **substitutions**: for $X, Y \subseteq_{\text{fin}} \text{Var}$ and for a map $\sigma: X \rightarrow \text{Term}(Y)$, we have a Boolean homomorphism $\mathbf{P}_\sigma: \mathbf{P}(X) \rightarrow \mathbf{P}(Y)$, modelling $\alpha \mapsto \alpha[(\sigma(x)/x)_{x \in X}]$,
3. **quantifiers**: for $X \subseteq_{\text{fin}} \text{Var}$, $y \in \text{Var} \setminus X$ two maps $\exists, \forall: \mathbf{P}(X \cup \{y\}) \rightarrow \mathbf{P}(X)$, modelling $\alpha \mapsto \exists y \alpha$ and $\alpha \mapsto \forall y \alpha$,

satisfying certain axioms (functoriality of substitutions, quantifiers are adjoint to adding dummy variables, quantifiers commute with substitutions).

We avoid equality, for simplicity.

First-order Boolean doctrines are the algebras of classical first-order logic.

Disclaimer: the “classical first-order logic” meant here is the one whose semantics allows the usage of empty structures.

If empty structures are undesired, one can restrict to the first-order Boolean doctrines satisfying “ $\exists x \top = \top$ ”.

The algebras of quantifier-free formulas modulo a universal theory $\rightsquigarrow$
Boolean doctrines.

These are defined as **first-order Boolean doctrines**, but without quantifiers: a family of Boolean algebras, linked by substitutions (Boolean homomorphisms), satisfying functoriality.

First-order Boolean doctrines over a fixed functional language $\mathcal{F}$ form a class of many-sorted algebras (one sort for each finite set of variables) definable by equations (i.e. a *variety of algebras*).

Varieties of algebras admit all free algebras. The forgetful functor from the category of **first-order Boolean doctrines** over $\mathcal{F}$ to the category of **Boolean doctrines** over $\mathcal{F}$ has a left adjoint: it maps a Boolean doctrine $\mathbf{P}$ to a first-order Boolean doctrine $\mathbf{P}^{\forall\exists}$, called the *quantifier completion* of $\mathbf{P}$, which “freely adds all quantifiers” to $\mathbf{P}$.

We have injections for each $X \subseteq_{\text{fin}} \text{Var}$

$$\mathbf{P}(X) \hookrightarrow \mathbf{P}^{\forall\exists}(X).$$

Roughly speaking: the set of quantifier-free formulas with variables in X is a subset of the set of first-order formulas with variables in X .

Herbrand's theorem

Let $\mathcal{T}$ be a universal theory in a language with at least a constant symbol. For $x \in \text{Var}$ and $\alpha(x)$ quantifier-free, $\mathcal{T}$ proves $\exists x \alpha(x)$ if and only if there are term-definable constants $c_1, \dots, c_n$ such that $\mathcal{T}$ proves $\alpha[c_1/x] \vee \dots \vee \alpha[c_n/x]$.

Algebraic version:

Theorem (A., Guffanti)

Let $\mathbf{P}$ be a Boolean doctrine over a functional signature. For $x \in \text{Var}$ and $\alpha \in \mathbf{P}(\{x\})$,

$$\exists x \alpha = \top \quad \text{in } \mathbf{P}^{\forall\exists}(\emptyset)$$

if and only if there are term-definable constants $c_1, \dots, c_n$ such that

$$\alpha[c_1/x] \vee \dots \vee \alpha[c_n/x] = \top \quad \text{in } \mathbf{P}(\emptyset).$$

No need for the existence of constants in $\mathcal{F}$. (Since $\emptyset$ is allowed.)

$\mathcal{T}$ cannot be an arbitrary theory $\rightsquigarrow$ we need freeness of $\mathbf{P}^{\forall\exists}$ over $\mathbf{P}$.

We actually proved it in the setting of first-order Boolean doctrines as certain functors $C^{\text{op}} \rightarrow \mathbf{BA}$.

In this way, we cover also *many-sorted* classical first-order logic.

The proof uses **models**. To produce them, we use a proof similar to Henkin's proof of Gödel's completeness theorem (using the axiom of choice).



Freely adding one layer of quantifiers to a Boolean doctrine. *arXiv link*.

Thank you!